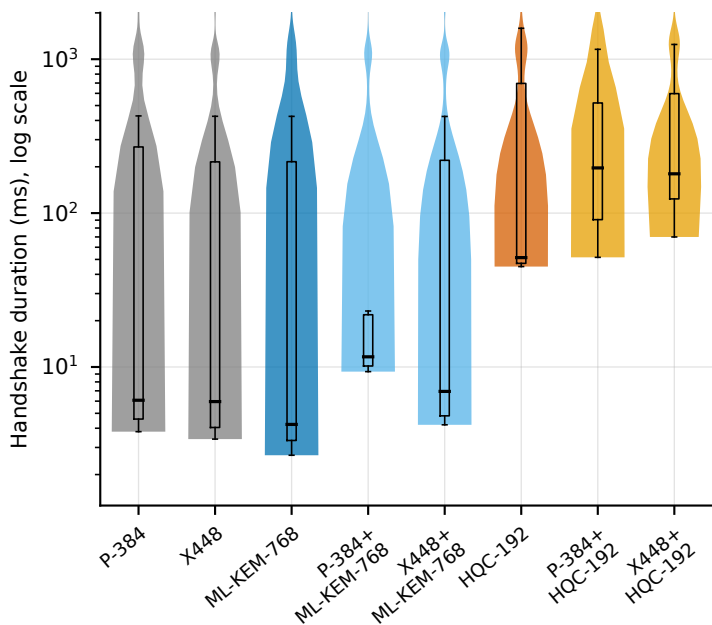


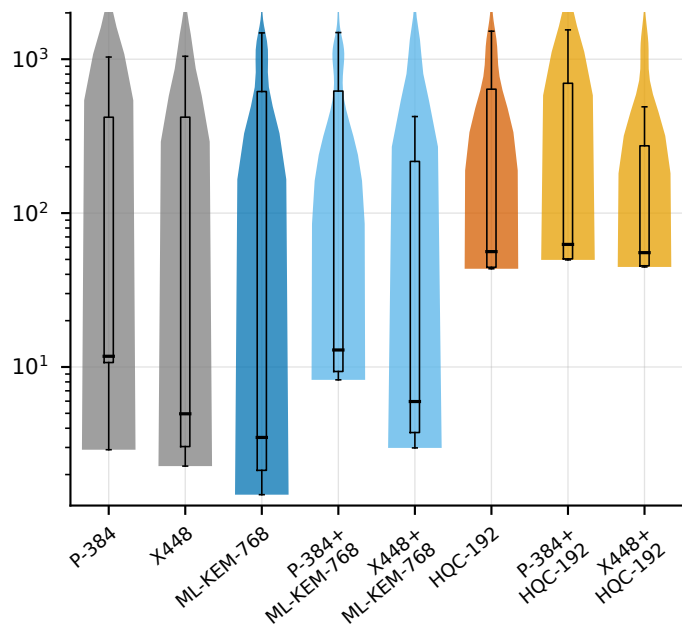
(a)

secp384r1 (classical)



(b)

mlDSA65 (post-quantum)



(c)

Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM              | secp384r1<br>Mean (ms) | secp384r1<br>CV (%) | secp384r1<br>Max (ms) | mlDSA65<br>Mean (ms) | mlDSA65<br>CV (%) | mlDSA65<br>Max (ms) |
|------------------|------------------------|---------------------|-----------------------|----------------------|-------------------|---------------------|
| P-384            | 330.9                  | 252.1               | 13.33k                | 487.4                | 510.7             | 53.25k              |
| X448             | 284.0                  | 241.9               | 9.70k                 | 439.2                | 328.5             | 28.57k              |
| ML-KEM-768       | 359.7                  | 332.3               | 14.17k                | 420.7                | 255.9             | 16.26k              |
| P-384+ML-KEM-768 | 266.6                  | 234.2               | 7.22k                 | 397.8                | 210.4             | 7.63k               |
| X448+ML-KEM-768  | 296.7                  | 209.8               | 4.55k                 | 382.1                | 347.1             | 26.41k              |
| HQC-192          | 456.4                  | 158.1               | 6.63k                 | 511.2                | 222.7             | 14.47k              |
| P-384+HQC-192    | 518.9                  | 280.7               | 29.94k                | 581.3                | 421.7             | 52.78k              |
| X448+HQC-192     | 524.8                  | 140.6               | 7.80k                 | 412.0                | 237.2             | 13.57k              |